

Direct Solution-Based Synthesis of $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ Solid Electrolyte

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All-solid-state batteries (ASSBs) promise higher power and energy density than batteries based on liquid electrolytes. Recently, a stable 3 V ASSB based on the super ionic conductor (1 mS cm^{-1} near room temperature) $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ has demonstrated excellent cycling stability. This study concerns the development of a five-step, scalable, and solution-based synthesis of $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$. The use of a wet chemistry approach allows solution processing with high throughput and

addresses the main drawbacks for this technology, specifically, the limited electrode–electrolyte contact and high cost. Moreover, a cost-efficient synthesis of the expensive precursors $\text{Na}_2\text{B}_{10}\text{H}_{10}$ and $\text{Na}_2\text{B}_{12}\text{H}_{12}$ is also achieved through the same process. The mechanism of the reactions is investigated and two key parameters to tune the kinetics and selectivity are highlighted: the choice of counter cation (tetraethylammonium) and solvent.

Introduction

All-solid-state batteries (ASSBs) are considered as promising electrochemical energy storage devices with increased energy and power density and improved safety.^[1] The development of ASSBs requires the challenges related to the solid electrolyte (SE) properties, notably, ionic conductivity and stability, as well as its integration into devices to be simultaneously addressed, including the chemomechanical stability of the electrode–electrolyte interfaces upon cycling.^[2] Cost-effective synthesis and processing methods are also required for future large-scale production of ASSBs.^[3] Significant efforts have already been made to develop materials with high ionic conductivity,^[4] and aspects related to device integration are becoming increasingly important.

Recently, an emerging class of materials, namely, *closo*-hydroborates,^[5] have demonstrated excellent performances in

this regard. $\text{Na}_2(\text{CB}_{11}\text{H}_{12})(\text{CB}_9\text{H}_{10})$ exhibits one of the highest ionic conductivity recorded (70 mS cm^{-1} at RT).^[4b] Duchêne et al. reported an operational all-solid-state 3 V stable battery based on the $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ SE, which possesses an ionic conductivity of approximately 1 mS cm^{-1} at 20°C and is compatible with a metallic sodium anode,^[5c, 6] whereas most other SEs are unstable toward metallic Na.^[7] The peculiar anion dynamic within $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ was identified to be responsible for the high cation diffusion.^[8] In addition, our group studied the behavior of $\text{Na}_2\text{B}_{12}\text{H}_{12}$ under pressure and demonstrated its high compressibility with a bulk modulus as low as 14 GPa.^[9] This result highlights that this class of materials could be suitable to accommodate volume changes of the electrodes during cycling. For the aforementioned reasons, *closo*-hydroborates-based electrolytes are promising candidates for the development of ASSBs technology. Although *closo*-hydroborates are solution processable,^[4b, 6, 10] they are often synthesized from toxic and expensive boranes (BH_3 , B_2H_6 or $\text{B}_{10}\text{H}_{14}$),^[11] hindering their development for any practical applications. Some syntheses starting from sodium tetrahydroborate (NaBH_4) in diglyme and iodine to oxidize NaBH_4 into NaB_3H_8 have been developed, however the purity and solvent removal are issues for the performance of the solid-state conductor.^[12]

Herein, we report a novel strategy to simultaneously address the cost effectiveness and processing challenges for *closo*-hydroborate SEs. We demonstrate a straightforward and selective wet chemistry approach to synthesize the superionic conductor and highly stable electrolyte $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$. Specifically, we present a five-step scalable synthesis starting from inexpensive, readily available and nontoxic precursors under mild conditions. The use of solvothermal synthesis instead of the ceramic routes that are usually used enable the design of more sustainable batteries.^[13] The process involves two intermediates of interest for hydroborate chemistry, namely B_2H_7^- and

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 Supporting Information, including all syntheses details, Nyquist plots, thermochemical data, additional NMR, FTIR and XRD, and a picture of the autoclave, and the ORCID identification number(s) for the author(s) of this article can be found under:
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$B_3H_8^-$.^[14] The final product is obtained by drying an isopropanol solution. Impedance spectroscopy of the as-prepared SE confirmed its high conductivity, which was independent from the synthesis route (ball milling versus wet chemistry). Finally, we discuss the mechanism of the reaction pathways and show the possibility to obtain deuterium-labeled samples under mild conditions.

Results and Discussion

Synthesis

Because of better control of the reaction steps, a solution-based method is often superior to the gas-solid or solid-state approaches that are often used for the synthesis of *closo*-hydroborates.^[11, 15] This strategy allows high throughput, facilitating scale-up to enhance productivity. To keep the process safe and to maximize cost competitiveness, $NaBH_4$ was used as a starting reagent. As $NaBH_4$ is poorly soluble in organic solvents and readily reacts with protic solvents such as alcohols, the first step was to exchange the inorganic cation (Na^+) with an organic one, in this case to form tetraethylammonium tetrahydroborate $[(C_2H_5)_4NBH_4, TEABH_4]$. The metathesis was achieved mechanochemically using a planetary ball-mill. The successful quantitative exchange was confirmed by FTIR spectroscopy, as shown in Figure 1. The B–H stretching modes characteristic of $NaBH_4$ ($2000\text{--}2600\text{ cm}^{-1}$) were not observed, whereas, a new signal in this region characteristic of $TEABH_4$ was observed, which was in agreement with a previous report.^[16]

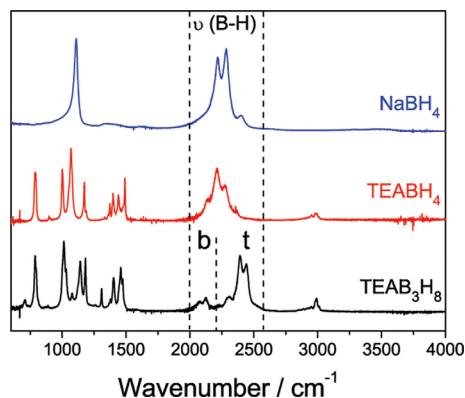


Figure 1. FTIR spectra of the $NaBH_4$ precursor (blue line), the $TEABH_4$ intermediate (red line) and the $TEAB_3H_8$ product (black line). The letters “b” and “t” indicate bridging and terminal B–H stretching modes, respectively.

For the second step towards larger boron clusters, tetraethylammonium octahydrotriborate $[(C_2H_5)_4NB_3H_8, TEAB_3H_8]$ was synthesized as an intermediate, following a previously developed strategy.^[10b] $TEABH_4$ was dissolved in dichloromethane (CH_2Cl_2 , DCM) and allowed to react with DCM by heating the mixture at $70^\circ C$ for 36 h in an autoclave. Typical terminal ($2250\text{--}2500\text{ cm}^{-1}$) and bridging ($2000\text{--}2200\text{ cm}^{-1}$) B–H stretching modes were observed (Figure 1),^[10b, 16, 17] confirming the formation of $TEAB_3H_8$.

Inspired by older studies,^[18] and following strategies developed in our previous work,^[10b] we discovered that thermal dehydrocoupling of $TEAB_3H_8$ in toluene at $185^\circ C$ in a closed system (step 3) leads to the selective formation of an equimolar mixture of $TEA_2B_{10}H_{10}$ and $TEA_2B_{12}H_{12}$ (Figure 2). The as-ob-

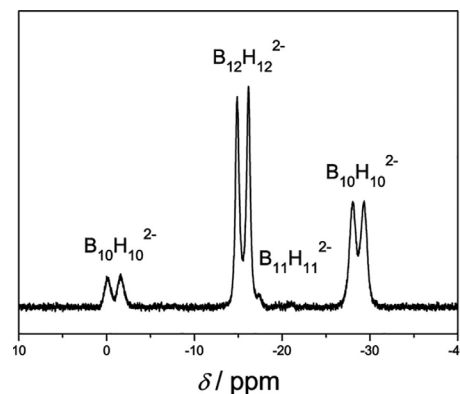
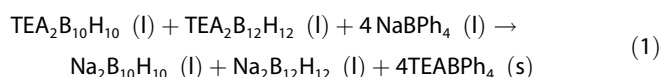
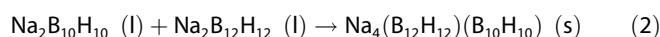


Figure 2. ^{11}B NMR spectrum recorded in $[D_6]DMSO$ of the solid phase obtained after 16 h thermolysis of $TEAB_3H_8$ at $185^\circ C$.

tained products can be processed in two different ways. The first approach was used to target the direct synthesis of $Na_4(B_{12}H_{12})(B_{10}H_{10})$ SE. To exchange the organic TEA cation back to sodium, we exploited the low solubility of tetraethylammonium tetraphenylborate $[(C_2H_5)_4NB(C_6H_5)_4, TEABPh_4]$ as a reaction product in alcohol or water by adding sodium tetraphenylborate $[NaB(C_6H_5)_4, NaBPh_4]$ in an equimolar mixture. This enabled the fast precipitation of $TEABPh_4$, which can be easily filtered out.^[19] The metathesis was performed in boiling water, in which both starting compounds are soluble, allowing the precipitation of $TEABPh_4$ as described by Equation (1).



Drying the solution of $Na_2B_{10}H_{10}$ and $Na_2B_{12}H_{12}$ did not lead to the formation of the mixed-anion compound $Na_4(B_{12}H_{12})(B_{10}H_{10})$ but rather to a mixture of the two individual materials $Na_2B_{10}H_{10}$ and $Na_2B_{12}H_{12}$ (step 4). We attributed this to hydrate formation, preventing the crystallization of the desired composition of $Na_4(B_{12}H_{12})(B_{10}H_{10})$. To circumvent hydrate formation, we dissolved the dried mixture of $Na_2B_{10}H_{10}$ and $Na_2B_{12}H_{12}$ (ratio 1:1) in isopropanol (IPA), which is less coordinating than water. After solvent removal, we obtained the electrolyte with the right defined composition [Eq. (2)] and the conductive fcc packing, as shown in Figure 3a (step 5).



The conductivity of the solution-processed electrolyte was in the order of 1 mS cm^{-1} near RT (Figure 3b and Figure S1 in the Supporting Information), comparable with the one previously reported for the same phase prepared through solid-state mechanical milling followed by heat treatment.^[5c] Therefore, our

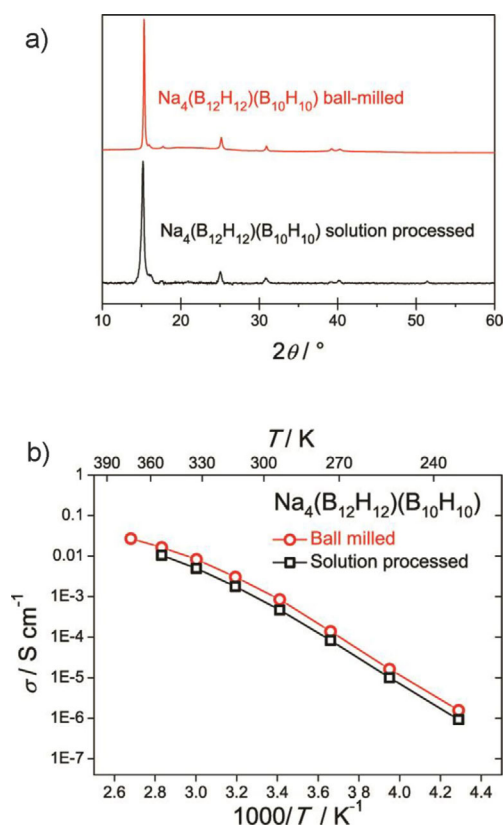
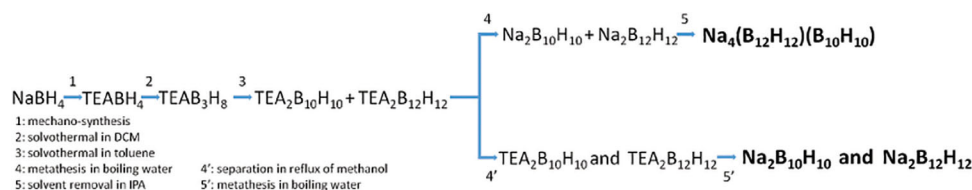


Figure 3. (a) Comparison of the XRD diffraction patterns of the products obtained through the wet synthesis approach (black line) and the mechanochemical route (red line). (b) Temperature-dependent Na^+ ion conductivity upon heating of the solution-processed $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ electrolyte compared with that of previous mechanochemically processed $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ electrolyte reported in the literature.^[5d]

synthesis method is particularly interesting in view of the scalable fabrication of ASSBs using solution processing methods.

The second approach was used to separate the equimolar mixture $\text{TEA}_2\text{B}_{12}\text{H}_{12} + \text{TEA}_2\text{B}_{10}\text{H}_{10}$ (obtained in step 3) to obtain the individual *closo*-hydroborates $\text{Na}_2\text{B}_{12}\text{H}_{12}$ and $\text{Na}_2\text{B}_{10}\text{H}_{10}$. This was achieved under reflux in methanol followed by fractional recrystallization of the liquid phase ($\text{TEA}_2\text{B}_{12}\text{H}_{12}$ being less soluble in methanol than $\text{TEA}_2\text{B}_{10}\text{H}_{10}$; step 4', Figure S2). Finally, we used the same aforementioned method for the salt metathesis in boiling water and obtained both sodium salts, $\text{Na}_2\text{B}_{12}\text{H}_{12}$ and $\text{Na}_2\text{B}_{10}\text{H}_{10}$, separately (step 5', Figure S3). Both salts are also valuable for basic research on this class of materials.

The whole process can be summarized as describe in Scheme 1, starting with NaBH_4 we directly obtained the SE



Scheme 1. Summary of the five-step process starting from NaBH_4 to the highly conductive electrolyte $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ and to $\text{Na}_2\text{B}_{12}\text{H}_{12}$ and $\text{Na}_2\text{B}_{10}\text{H}_{10}$ separately.

$\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ through a 5 steps process and demonstrated the possibility to separate both *closo*-hydroborates. Except for the use of DCM, the solvents used in this process are alcohol, water and toluene; the latter can be produced from bioethanol,^[20] which adheres to a sustainable development approach. Moreover, we believe that the use of DCM can be reduced to the role of reactant using stoichiometric quantities or even exchanged by another electrophilic agent.

Mechanism

TEAB₃H₈

To monitor the second reaction step, different reaction times (1, 3, 8, 12, 16, 24, and 36 h) were chosen and the progress of this solvothermal synthesis was monitored by ex situ ^{11}B NMR spectroscopy (Figure 4). The complete conversion of BH_4^- into B_3H_8^- was achieved within 36 h at 70°C . The signal corresponding to B_2H_7^- confirmed the mechanisms proposed by B rchner et al. and our group for the formation of tetrabutylammonium octahydrotriborate (TBAB_3H_8).^[10b, 16, 21] In this experiment, the conversion of BH_4^- into B_2H_7^- was more obvious than in previous studies, revealing that the reaction involves two steps, that is, $\text{BH}_4^- \rightarrow \text{B}_2\text{H}_7^- \rightarrow \text{B}_3\text{H}_8^-$. Between 8 and 12 h only B_2H_7^- is formed. After 12 h B_2H_7^- start to convert into B_3H_8^- and is completely converted after 36 h. These observations differ from those made in previous studies for the synthesis of TBAB_3H_8 in two ways: First, for TBAB_3H_8 , the formation of a borane adduct [tributylborane (TrBu-BH_3)] has been reported and proposed to be involved in the mechanism, and

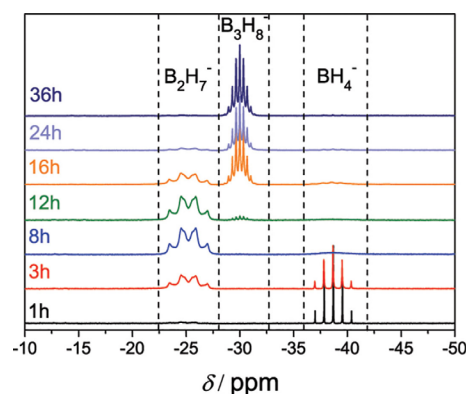
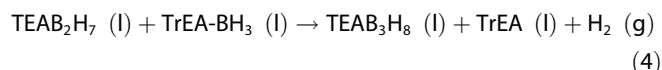


Figure 4. ^{11}B NMR spectra of the liquid phase during the solvothermal synthesis of TEAB_3H_8 at different reaction times. NMR experiments were performed in CD_2Cl_2 .

second, the signal corresponding to the formation of TBAB_2H_7 had very low intensity compared with the signal of TEAB_2H_7 recorded herein. From these observations, we conclude that the adduct triethylamine borane (TrEA-BH_3) is more reactive than TrBA-BH_3 , which reacts readily with TEABH_4 to give TEAB_2H_7 [Eq. (3)] whereas, TEAB_2H_7 is more stable than TBAB_2H_7 , lowering the kinetics of the conversion into TEAB_3H_8 [Eq. (4)].



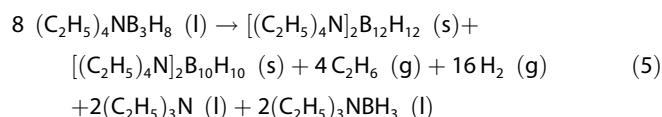
From this conclusion, the choice of the counter cation is a crucial parameter to selectively obtain the desired product, confirming that solution-based routes allow tuning of the reaction mechanism. Finally, this procedure can be scaled up to at least 10 g scale (see synthesis details in the Supporting Information and Figure S4).

To gain an insight into the role of the B–H bond reactivity, we performed the synthesis starting from the deuterated precursor, NaBD_4 . After 36 h, a majority of B_2D_7^- was formed and increasing the temperature to 90°C was necessary to convert B_2D_7^- into B_3D_8^- (Figure S5). This result demonstrates that the B–H bond breaking to form hydrogen described in Equation (4) is the rate limiting step. Moreover, through this experiment, we demonstrate that the procedure can be adapted to synthesize deuterated B_3D_8^- and $\text{B}_2\text{D}_7^{2-}$ salts in a liquid phase starting from NaBD_4 , which is of high interest for neutron diffraction studies.

$\text{TEA}_2\text{B}_{10}\text{H}_{10}$ and $\text{TEA}_2\text{B}_{12}\text{H}_{12}$

To identify the different byproducts after dehydrocoupling of TEAB_3H_8 , we performed 1D ^1H (Figure S6) and ^{11}B (Figure S7) NMR on the liquid and the solid obtained after 16 h at 185°C and 2D ^1H – ^{13}C heteronuclear single quantum coherence (HSQC) NMR on the liquid phase (Figure S8). This allowed the identification (see experimental section for NMR assignments) of the selective formation of $\text{TEA}_2\text{B}_{10}\text{H}_{10}$ and $\text{TEA}_2\text{B}_{12}\text{H}_{12}$ with a ratio 1:1 (obtained by integration of the ^{11}B NMR spectrum) and $\text{TEA}_2\text{B}_{11}\text{H}_{11}$ as a minor impurity in the solid phase (Figure 2).

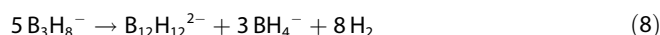
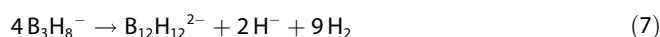
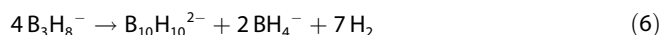
With regards to the liquid phase, we identified the formation of triethylamine borane [$(\text{C}_2\text{H}_5)_3\text{NBH}_3$, TrEA-BH_3 ; Figures S5, S6, and S7] and triethylamine [$(\text{C}_2\text{H}_5)_3\text{N}$, TrEA ; Figures S6 and S7].^[22] Based on these observations, we propose the following equation for the reaction [Eq. (5)]:



Makhlouf et al. performed a similar study on the thermolysis of TEAB_3H_8 in dry conditions at 185°C , yet they did not observe the formation of TrEA-BH_3 and $\text{TEA}_2\text{B}_{12}\text{H}_{12}$.^[18] They also

found that dry thermolysis at 185°C starting with tetramethylamine octahydrotriborate [$(\text{CH}_3)_4\text{B}_3\text{H}_8$, TMAB_3H_8], leads to the formation of similar products as describe by Equation (5), although no TrEA was detected. Furthermore, despite our efforts, the thermal treatment of TBAB_3H_8 in toluene did not lead to the formation of any *clos*o-hydroborates. Consequently, the cation, as previously noticed, and the solvent influence the reaction mechanism by stabilizing the intermediates during the addition of boron in the growth of the *clos*o-hydroborates cluster. As a result, TrMA-BH_3 , TrEA-BH_3 and TrBA-BH_3 have different reactivity leading to different kinetics and selectivity, and finally different products are obtained. In other words, we identified two parameters that can be used to tune the reaction, the cation and the solvent.

Reaction 5 implies that the formation of $\text{B}_{10}\text{H}_{10}^{2-}$ and $\text{B}_{12}\text{H}_{12}^{2-}$ is entropy-driven owing to the release of 16 hydrogen molecules. We recently calculated the formation enthalpies of neutral and charged tetrahydroborates in the gas phase by density functional theory (DFT),^[23] as well as their temperature-dependent thermodynamic properties from anharmonic DFT calculations.^[24] Using these data in conjunction with thermodynamic data for H^- and H_2 , we computed the free energy of the following reactions in the gas phase as a function of temperature [Eq.(6)–(8)]:



The reactions in Equations (6) and (7) describe together the simplified reaction of $8 \text{B}_3\text{H}_8^-$ in Equation (5). Both reactions become spontaneous at 500 K, in line with the experimental observations, and qualitatively explain the formation of both $\text{B}_{10}\text{H}_{10}^{2-}$ and $\text{B}_{12}\text{H}_{12}^{2-}$, however, without considering the contribution of the counterions, which are experimentally very important. Interestingly, the reaction in Equation (8), which is in principle also possible, is spontaneous even at 300 K with $\Delta_r G^\circ = -228.6 \text{ kJ mol}^{-1}$. In this case, the progress of the reaction must be kinetically controlled, as no reaction was observed at 300 K

Conclusions

Two significant challenges regarding the development of ASSBs technology, that is, the processing of SEs and their expensive synthesis, were addressed. The $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ *clos*o-hydroborate-based SE with the right composition and without loss of conductivity was selectively obtained from solution. This allows the use of a conventional slurry coating technique to obtain sheet-like electrodes with high electrode/electrolyte ionic contact.^[6,25] Furthermore, the inherent recrystallization with IPA allows a consistent purity of the electrolyte to be obtained, which is not guaranteed when ball-milling solids of different origin.

A cost-effective process was developed starting with the inexpensive and safest hydroborate source available: NaBH_4 . A solution-based approach allowed easy scale-up of the process. Moreover, most of the solvents used during the synthesis, especially water and IPA, are environmentally friendly, cost effective and readily available in laboratories. Toluene can be obtained from biomass; however, owing to its toxicity it would be better to use another solvent such as limonene. DCM could be used as reactant in stoichiometric quantities, which would significantly decrease the required amount. The reaction mechanism of the different steps was proposed, which highlighted the importance of the solvent and the nature of the organic cation in tuning the kinetics and selectivity. The solvothermal method described herein enables the screening of more solvents at temperatures above their boiling point to improve the sustainability and the cost effectiveness of the process. Moreover, the potential emission of toxic boranes such as diborane B_2H_6 is confined in the autoclave, preventing safety issues and loss of the boron source, which would be detrimental for the yield of the synthesis. Therefore, this process paves the way for development of a large-scale synthesis for further studies and the design of novel all-solid-state sodium batteries.

Experimental Section

Characterizations

FTIR experiments were performed with a Biorad Excalibur Instrument equipped with a portable Specac Golden Gate heatable ATR setup, which allows the preparation of the samples in the glovebox. IR spectra were recorded with a spectral resolution of 1 cm^{-1} , with 30 scans in the range $600\text{--}4000\text{ cm}^{-1}$.

^1H and HSQC NMR experiments were performed at 298 K using a Bruker Avance III 500 MHz spectrometer with a 5-mm DCH $^{13}\text{C}\text{--}^1\text{H}$ /D helium-cooled cryogenic probe equipped with a z-gradient coil. 2D HSQC experiments (Bruker pulse program "hsqcetgpsisp2.2") were recorded in 5 min (for commercial samples) and in 40 min (for experimental samples) with a 17 ppm spectral width in the ^1H dimension and a 200 ppm spectral width in the ^{13}C dimension. Each of the 256 increments in t_1 was acquired with 1 scan (for commercial samples) and with 8 scans (for experimental samples), a 1.0 s recycle delay between scans and 2048 points in the direct dimension.

For ^{11}B NMR spectra, two spectrometers were used: $\text{TEA}_2\text{B}_{12}\text{H}_{12}$ and $\text{TEA}_2\text{B}_{10}\text{H}_{10}$ spectra were recorded at 298 K on a Bruker ADVANCE III HD-NanoBay 400 MHz spectrometer, equipped with a 5 mm CryoProbe Prodigy (CPPBBO BB-1H/19F/D), at a frequency of 400.13 MHz. TEAB_3H_8 experiments were performed at RT on a Bruker ADVANCE III HD-NanoBay 300 MHz spectrometer, equipped with a 5 mm BB(F)-H-D probe, at a frequency of 300.13 MHz.

Powder XRD data were collected in a Debye-Scherrer geometry (0.5 mm diameter sealed capillary) on a STOE diffractometer STADI-P with a 1D plate IP-PSD detector and a $\text{CuK}_{\alpha 1}$ radiation ($\lambda = 1.5406\text{ \AA}$) with a germanium monochromator.

Ionic conductivity was measured by electrochemical impedance spectroscopy using a BioLogic VMP3 multichannel potentiostat in a frequency range from 1 MHz to 1 Hz with a voltage amplitude of 10 mV. The powder samples were pressed under a pressure of 500 MPa into a 10 mm diameter pellet with a thickness of 0.85 mm. The pellet was loaded in a custom-made air-tight cell and contacted between two spring loaded electrodes. Indium foils

were mechanically pressed on the pellet surfaces to improve the contact between the sample and the molybdenum electrodes. Ionic conductivity was measured in steps of 20°C between -40°C and 80°C . The temperature was controlled by a climatic chamber and equilibrating the cell for 2 h at each temperature.

NMR Assignments

^1H NMR

NMR characterization of the liquid phase after thermolysis of TEAB_3H_8 at 185°C during 16 h: TrEA-BH_3 : $\delta = 0.84$ (t, $J = 7.3\text{ Hz}$, $-\text{CH}_3$) and $\delta = 2.32\text{ ppm}$ (q, $J = 7.3\text{ Hz}$, $-\text{CH}_2-$), TrEA : $\delta = 1.00$ and $\delta = 2.41\text{ ppm}$, signal too broad to distinguish coupling.

NMR characterization of commercial samples: TrEA-BH_3 : $\delta = 0.88$ (t, $J = 7.3\text{ Hz}$, $-\text{CH}_3$), $\delta = 2.37\text{ ppm}$ (q, $J = 7.3\text{ Hz}$, $-\text{CH}_2-$). TEA : $\delta = 2.42$ (q, $J = 7.20\text{ Hz}$, $-\text{CH}_2-$), $\delta = 1.00\text{ ppm}$ (t, $J = 7.20\text{ Hz}$, $-\text{CH}_3$).

^{11}B NMR

BH_4^- : $\delta_{11\text{B}} = -39\text{ ppm}$ (q, $^1J_{\text{B,H}} = 82\text{ Hz}$), B_2H_7^- : $\delta_{11\text{B}} = -25.3\text{ ppm}$ (quartet of doublets), B_3H_8^- : $\delta_{11\text{B}} = -30.1\text{ ppm}$, (n, $^1J_{\text{B,H}} = 33\text{ Hz}$), $\text{B}_{12}\text{H}_{12}^{2-}$: $\delta_{11\text{B}} = -15.5\text{ ppm}$ (d, $^1J_{\text{B,H}} = 125\text{ Hz}$), $\text{B}_{10}\text{H}_{10}^{2-}$: $\delta_{11\text{B}} = -28.8$ (d, $^1J_{\text{B,H}} = 125\text{ Hz}$, 8B, 2 to 9) and -0.6 ppm (d, $^1J_{\text{B,H}} = 141\text{ Hz}$, 2B, 1 and 10, see Figure S8 for numbering), $\text{B}_{11}\text{H}_{11}^{2-}$: $\delta_{11\text{B}} = -17.2\text{ ppm}$ doublet overlapped with $\text{B}_{12}\text{H}_{12}^{2-}$, TrEA-BH_3 : $\delta_{11\text{B}} = -12.3\text{ ppm}$ (q, $^1J_{\text{B,H}} = 97\text{ Hz}$).

HSQC NMR

TrEA : $-\text{CH}_2-$ signal at 2.41/46.5 ppm, $-\text{CH}_3-$ signal at 1.00/12.4 ppm, TrEA-BH_3 : $-\text{CH}_2-$ signal at 2.32/52.0 ppm, $-\text{CH}_3-$ signal at 0.84/8.1 ppm.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: batteries • boron • energy storage • solid electrolytes • solid-state structures

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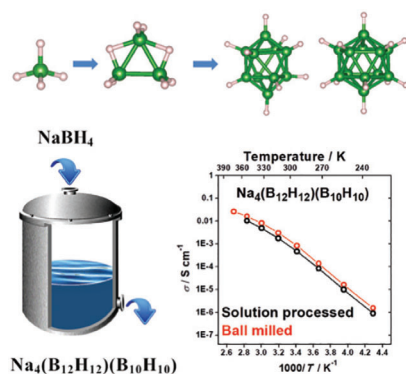
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FULL PAPERS

Low cost, low toxicity, high five! A facile five-step solution-based synthesis of the superionic conductor $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ was developed starting from the low cost and safe hydroborate NaBH_4 without the use of toxic boranes. The synthesis method is a step forward for the design of novel all-solid-state batteries using impregnation techniques with high throughput.



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Direct Solution-Based Synthesis of $\text{Na}_4(\text{B}_{12}\text{H}_{12})(\text{B}_{10}\text{H}_{10})$ Solid Electrolyte

